Emotion Recognition from Speech and Text using a Multimodal Transformer with Emotionally Salient-Aware Positional Encoding and Cross Attention Mechanism

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Abstract

Accurate understanding of human emotions from speech is essential in developing emotionally intelligent systems which can facilitate applications like digital therapy, human-computer interaction, and AI agents. The unimodal systems tend to miss the complete range of emotional cues because processing either audio or text has its own limitations. This encourages the construction of a more complete multimodal approach that has a better ability to recognize emotion by considering both what is uttered and how it is uttered. In this paper, we introduce a multimodal speech emotion recognition system that takes advantage of both the audio and textual modalities. The system includes two parallel deep learning models, a voice model and a text model, each individually optimized for modality-specific features. The voice-centric model handles temporal frame-level features (e.g., MFCCs, pitch, formants, and spectral features) as well as non-temporal global features (such as jitter, shimmer, and speech rate). It utilizes a 1D convolutional frontend and a new Emotionally Salient-Aware Positional Encoding (ESaPE) approach, which involves multi-scale encodings in terms of frames, syllables, and phonemes. This is complemented by a gated fusion between self and cross-attention mechanisms, combining global statistical information with local acoustic dynamics. Concurrently, the text-based model learns to fine-tune a DistilBERT transformer on transcribed speech to bring out semantic and contextual information pertaining to emotional states. The forecasts from the two modalities

are fused together employing multiple late fusion approaches, making it possible for the system to take advantage of the complementary benefits of voice and text. The resultant framework is shown to provide better robustness and emotional comprehension across different speech situations, representing a step higher in the evolution of empathetic AI .

Keywords: Speech Emotion Recognition, Emotionally Salient Positional Encoding, Multi-modal Computing, Speech processing, Natural Language Processing, Acoustic features, Phonetic features, Deep Learning

1 Introduction

Emotions are at the heart of how we connect and communicate as humans, and often, it's not just what we say, but how we say it that truly conveys how we feel. The tone of voice, the pauses, the rhythm, all of these subtle cues carry emotional information that goes beyond words. This is where Speech Emotion Recognition (SER) comes into play. It's a field focused on teaching machines to understand human emotions from speech, with powerful applications in areas like empathetic virtual assistants, smarter customer service, and even mental health support.

Despite the growing interest in SER, building systems that can accurately recognize emotions from voice remains a tough challenge. Human speech is incredibly nuanced. Traditional models typically focus on either temporal features, like pitch changes or spectral patterns that evolve over time, or on non-temporal, global features such as jitter, shimmer, or average speech rate. While each of these brings something to the table, relying on just one side of the picture often means missing the complex dynamics that truly make speech emotional.

In this research, we propose an enhanced Transformer-based architecture that brings the best of both worlds. Our model combines detailed frame-level (temporal) features with broader statistical (non-temporal) features to capture both fine-grained emotion shifts and overall vocal trends. To push this further, we propose an Emotionally Salient-Aware Positional Encoding (ESaPE) strategy. Instead of treating all parts of speech equally, this approach helps the model focus more on the emotionally rich segments of the audio, those syllables or phonemes that carry the most emotional weight. We also incorporate cross-attention between the two types of features, enabling the model to learn deeper emotional patterns across different timescales.

To measure the performance, we merge four widely used emotion datasets CREMA-D[4], MELD[6], RAVDESS[3], and TESS[5], into a single, diverse benchmark for the voice-based model. The results show clear improvements over baseline models, demonstrating how combining multi-scale features with emotionally aware temporal modeling can lead to more accurate emotion recognition.

2 Literature Survey

Speech emotion recognition has evolved from traditional signal processing methods to deep learning models, with the current state-of-the-art being Transformer-based

architectures. These models excel at handling long-range dependencies and contextual nuances in audio signals [1, 2]. A critical aspect of constructing emotion recognition systems is the selection of datasets, each offering unique features vital for training robust models.

The Ryerson Audio-Visual Database of Emotional Speech and Song (RAVDESS) is renowned for its balanced set of vocal performances across eight emotion classes, such as calm, happiness, sadness, anger, fear, surprise, and disgust, each expressed at two levels of emotional intensity (normal and strong), along with a neutral expression, delivered by professional voice actors. Its inclusion of both speech and song versions introduces diversity in prosodic and acoustic patterns [3]. Similarly, the CREMA-D dataset offers high speaker variability with over 90 actors expressing various emotional states, including: happy, sad, anger, fear, disgust, and neutral. Its real-world variability makes it suitable for studies in generalization [4]. The Toronto Emotional Speech Set (TESS) is another valuable resource, featuring female voice actors reciting semantically neutral sentences across seven emotions, aiding in isolating the effect of prosody over lexical semantics [5]. The Multimodal EmotionLines Dataset (MELD) extends emotion recognition to multi-speaker conversations, providing audio, textual, and visual cues over multi-turn dialogues, thus enabling contextual emotion analysis in realistic settings such as group conversations [6].

The effectiveness of emotion recognition systems heavily relies on the representation of the input speech signal. Traditionally, non-temporal features like Mel-Frequency Cepstral Coefficients (MFCCs), pitch, energy, and formants have been extracted, capturing short-term spectral speech features. While rich in phonetic information, these features fall short in capturing long-term dependencies and transitions essential for identifying emotions that evolve over time [7]. To address this limitation, temporal features like delta and delta-delta coefficients and raw waveform segments are employed to represent dynamic speech features. Recently, self-supervised and hybrid methods have gained popularity, utilizing spectrograms or log-Mel spectrograms fed through learnable frontends to facilitate end-to-end training [8].

These methods often employ temporal convolution layers, such as 1D CNNs, to extract hierarchical patterns from raw or preprocessed signals. In Transformer-based models, this Conv1D frontend is crucial for achieving localized representations and reducing temporal resolution before feeding into the attention mechanisms [1]. Positional encoding is another critical component, as Transformers lack native recurrence. Positional encodings, whether learned or sinusoidal, are added to input embeddings to provide a sense of order, allowing the model to identify sequential relationships within speech [2].

The core of the architecture, the Transformer encoder, utilizes self-attention to compute weights on the relative importance of each frame or segment concerning all others, enabling it to learn global dependencies [2]. This is particularly beneficial in emotion detection, where information can be dispersed over non-consecutive frames. Innovations such as cross-attention module design allow the model to attend to multiple streams, for instance, combining acoustic features with semantic information (e.g., transcriptions) or visual information (e.g., facial landmarks in MELD or CREMA-D) [6, 9]. The cross-attention mechanism enables the model to selectively

focus on relevant emotional content across modalities, leading to significant performance improvements in multimodal environments [9, 10]. Additionally, Transformer variants like the Conformer architecture have been proposed, combining convolutional modules with self-attention layers to learn both local and global dependencies in audio signals more effectively [1, 11]. These hybrids address the shortcomings of pure attention models in modeling fine-grained prosodic variations necessary for emotion recognition.

Several studies have confirmed the superiority of Transformer-based models over recurrent neural networks (RNNs) and convolutional models. For instance, models trained on the RAVDESS and CREMA-D datasets have shown that attention-based models outperform others in terms of generalization across speakers and emotional states, thanks to their dynamic receptive fields and ability to focus on areas of high emotional content [3, 4]. Models trained on MELD using cross-attention mechanisms between audio and text modalities significantly outperform unimodal baselines, highlighting the importance of contextual information in understanding emotions [6, 9]. In these models, temporal features extracted using Conv1D layers enable the Transformer to process compressed yet information-rich representations, and positional encoding ensures that sequential information along the signal is maintained [1, 2]. Pretraining on large speech corpora using self-supervised models such as wav2vec or HuBERT, followed by fine-tuning using Transformer backbones, has also proven highly effective [8, 12]. Combining these pretrained features with Transformer encoders and attention pooling facilitates effective transfer learning and enhances performance even in low-resource settings like TESS [5, 10].In the existing work, deep temporal multisource feature modeling has been explored, where Transformers effectively fuse audio, text, and visual cues to capture complex emotional patterns, enhancing accuracy and robustness in real-world conditions.

3 Methodology

In this work, we propose a multimodal speech emotion recognition system, as shown in Figure 1, that jointly leverages both audio and textual information to enhance emotion classification accuracy. Our approach consists of two parallel deep learning models: a voice-based model and a text-based model, each of which is trained on features unique to its type, taken from specialized datasets.[13]

The voice-based model processes both frame-level temporal features (e.g., MFCCs, ZCR, pitch, formants, chroma, spectral rolloff) and non-temporal features (e.g., jitter, shimmer, speech rate, HNR, pitch range, formant ratios). A 1D convolutional frontend captures local acoustic structures,[14].

It is then followed by an Emotionally Salient Frame-Wise Positional Encoding (ESaPE) strategy that fuses multi-scale positional encodings (frame, syllable, phoneme), each with its own learnable scale. The syllable and phoneme positions are computed by converting alignment timestamps into frame-wise index arrays and are further modulated by learnable salience weights, allowing the network to dynamically emphasize emotionally critical frames in the speech signal.

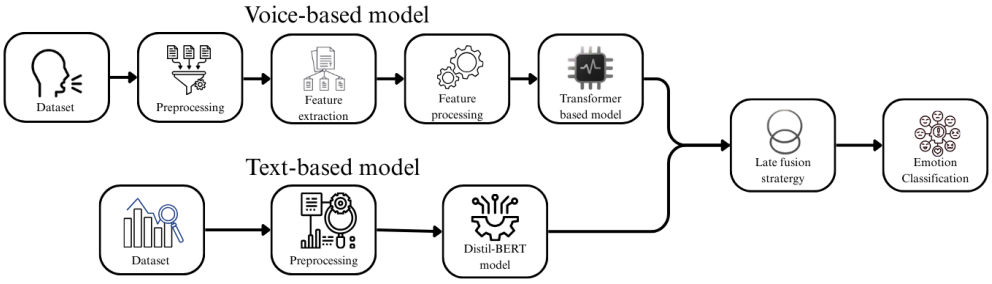


Fig. 1 Multimodal Emotion Recognition architecture

This encoding is added to the input before the Transformer layers. To effectively integrate local and global information of each utterance, a cross-attention mechanism correlates each frame with a projection of global (non-temporal) features. A gated fusion mechanism dynamically adjusts the contributions of self-attention and cross-attention across four Transformer blocks. The output is then added to the input of self-attention and normalized before passing through the feed-forward neural network, followed by another round of addition and normalization of the residuals.

Finally, the outputs are globally pooled and fused with projected global features before classification. This architecture facilitates a comprehensive understanding of both short-term sound patterns and broader emotional or prosodic context.

In parallel, the text-based model utilizes a pre-trained DistilBERT transformer, fine-tuned on emotion-annotated transcripts, to extract semantic and contextual cues from text[15]. The [CLS] token embedding is used to represent the utterance-level emotion state.

The outputs of both models are fused at the classification level using late fusion strategies, combining the predicted emotion probabilities from each modality to produce the final classification. This framework combines the strengths of speech and text to create a more accurate and robust emotion recognition system.

3.1 Voice-based model

3.1.1 Data Collection

In this work, we utilize a combination of four publicly available emotion recognition datasets to train and evaluate our voice-based emotion recognition model.

The need for a diverse representation of emotional expressions across multiple dimensions drives the choice of datasets. By incorporating both isolated speech samples and conversational data, the model is exposed to a broad range of emotional cues, ensuring its generalizability to real-world scenarios. The inclusion of both male and female speakers, as well as variations in speech tone, further strengthens the model's ability to detect emotions in diverse vocal expressions.

The subsequent datasets were incorporated into the work.

- CREMA-D (Crowd-sourced Emotional Multimodal Actors Dataset)

  - CREMA-D includes 7,442 clips from 91 actors and actresses, with diverse age and ethnicity, expressing six of the seven "universal emotions"[4].
  - Each clip is available in three modalities—audio-only, visual-only, and audio-visual—and was rated by 2,443 crowd-sourced annotators. Raters provided categorical emotion labels and intensity scores; majority voting was used to determine the group's perceived emotion.
  - The dataset offers high-quality, controlled audio recordings that contains multiple emotional expressions, making it ideal for training robust emotion recognition models in speech.

- MELD (Multimodal EmotionLines Dataset)
  - MELD is derived from TV show dialogues and contains over 13,000 utterances across 1,433 conversations [6].
  - The dataset includes a variety of emotional expressions, such as happiness, sadness, anger, fear, surprise, disgust, and neutrality.
  - While MELD includes multimodal data (text, audio, and visual features), we specifically focus on the audio components for emotion recognition. The conversational nature of this dataset introduces complexity, enriching the model's ability to recognize emotions in dynamic speech patterns.

- RAVDESS (Ryerson Audio-Visual Database of Emotional Speech and Song)
  - The RAVDESS contains 24 professional actors (12 female, 12 male), vocalizing two lexically-matched statements in a neutral North American accent[3].
  - The audio is highly controlled, ensuring high-quality recordings with minimal noise, making it suitable for training models focused on voice emotion recognition.

- TESS (Toronto emotional speech set)
  - TESS consists of 200 target words, each spoken within the carrier phrase "Say the word _" by two female actors aged 26 and 64. Each word was recorded while portraying one of seven emotional states: anger, disgust, fear, happiness, pleasant surprise, sadness, and neutral, resulting in a total of 2,800 audio samples[7].
  - The dataset is structured such that each actor's recordings are contained within separate directories, with subfolders corresponding to each emotion. Within these emotion-specific folders, the 200 target word utterances can be found.

To maximize the diversity of emotional speech patterns, the datasets were merged, resulting in a training set that contains over 20,000 labeled speech samples.

3.1.2 Data Preprocessing

We utilize four widely used emotional speech datasets: CREMA-D, TESS, RAVDESS, and MELD, each containing labeled utterances in diverse acoustic environments and speaking styles. To ensure uniformity across all datasets and prepare the data for multimodal analysis (text and audio), the following preprocessing steps were applied to a total of 21,671 utterances:

- **Audio Format Conversion.**

– As the audio data in all datasets (except MELD) was already provided in the (`.wav`) format, no format conversion was required for them.
– For the MELD dataset, which stores utterances in (`.mp4`) video format, we used the MoviePy library to extract and convert audio streams into (`.wav`) files.
– Figure 2 shows that most audio clips are short, with a right-skewed distribution. The mean duration is 2.84 seconds, while the median is 2.44 seconds, indicating the presence of some longer clips that raises the average.

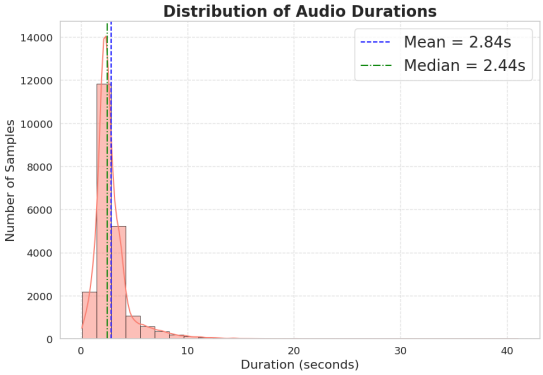


Fig. 2 Distribution of audio clip durations after format normalization.

- **Transcript Generation.**
  – For datasets lacking transcriptions (CREMA-D and TESS), we used OpenAI's Whisper model (medium) to generate high-quality speech-to-text transcriptions.
  – The RAVDESS dataset contains only two pre-defined scripted statements per utterance; these were directly mapped based on file naming conventions.
  – MELD provides its own transcriptions, which were directly used.
  – Figure 3 shows that transcript lengths are generally short, with a mean of 6.4 words per transcript. The distribution is right-skewed, indicating that most transcripts contain only a few words, while a small number extend significantly longer.

  All resulting transcriptions were stored in a new column labeled Transcript, ensuring textual modality was available for all datasets.
- **Label Harmonization.**
  – Emotion labels varied across datasets in both granularity and naming. A normalization dictionary was created to standardize these to a unified set: `{anger, sadness, fear, happy, disgusted, neutral, surprise, calm, joy}`.
  – Figure 4 illustrates the class distribution across emotion categories in the voice-based model. The data is imbalanced, with the 'Neutral' class making up the largest portion at 29%, while 'Calm' is significantly underrepresented at less than 1%.

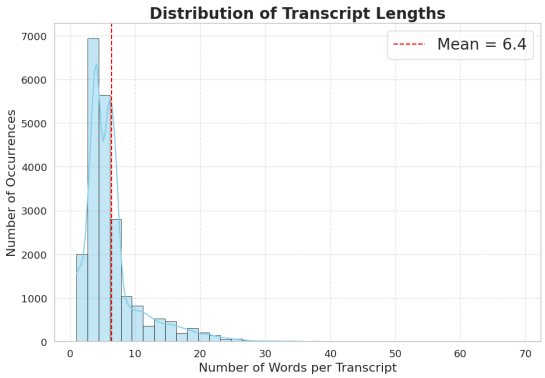


Fig. 3 Distribution of transcribed utterance lengths across merged dataset

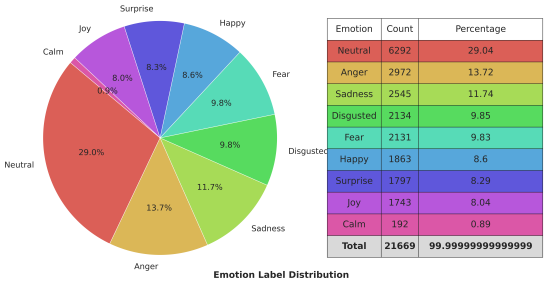


Fig. 4 Class distribution in voice-based model across multiple emotion categories.

- **Phoneme and Syllable boundaries generation.**
  To obtain precise phoneme and syllable boundaries with annotations, we utilized the Montreal Forced Aligner (MFA)[16].

  1. **Corpus Preparation and Forced Alignment**
     Each utterance was organized into a speaker-specific directory containing the corresponding audio (`.wav`) and transcript (`.txt`) files. Each utterance u_i in the dataset is represented by the pair

     $$u_i = \left(x_i(t), \texttt{Transcript}_i\right),$$

     where $x_i(t)$ is the raw audio waveform (sampled at 16 kHz) and $\texttt{Transcript}_i$ is the corresponding text file. For each utterance:

     – A speaker-specific directory $\texttt{mfa_corpus/speaker}_i$ is created.
     – The audio $x_i(t)$ is saved as $\texttt{utt}_i\texttt{.wav}$ within that directory.
     – The transcript $\texttt{Transcript}_i$ is saved as $\texttt{utt}_i\texttt{.txt}$ in the same folder.

     These files were processed using MFA, generating (`.TextGrid`) files containing aligned phoneme annotations.
  2. **Phoneme Boundary Extraction**

For each `.TextGrid` file corresponding to utterance u_i, we locate the tier named *phones*, which contains a sequence of phoneme intervals:

$$\text{Tier}_{\text{phones}} = \left\{(p_{i,j},\, t_{i,j}^{\text{start}},\, t_{i,j}^{\text{end}})\right\}_{j=1}^{n_i}.$$

Here:

– $p_{i,j}$ is the j-th phoneme symbol (e.g., `AA1`, `SH`, etc.).
– $t_{i,j}^{\text{start}}$ and $t_{i,j}^{\text{end}}$ denote its start and end times (in seconds), as returned by MFA.

We then apply a regular-expression-based cleaning step to remove stress markers (digits) and any non-phonetic symbols. Denote the cleaned phoneme labels by

$$\widetilde{p}_{i,j} = \text{clean}\big(p_{i,j}\big),$$

where clean$(\cdot)$ strips out digits and normalizes to lowercase. The resulting phoneme set for u_i is

$$\Pi_i \;=\; \left\{\big(\widetilde{p}_{i,j},\, t_{i,j}^{\text{start}},\, t_{i,j}^{\text{end}}\big)\right\}_{j=1}^{n_i}.$$

3. **Sonority Hierarchy and Vowel Set**
To group phonemes into syllables, we define a discrete sonority function

$$s : \mathcal{P} \;\to\; \mathbb{Z},$$

mapping each cleaned phoneme symbol $\widetilde{p}$ to an integer score according to the following hierarchy:

$$s(\widetilde{p}) = \begin{cases} 10, & \text{if } \widetilde{p} \in \{\texttt{a,e,i,o,u}\} \quad \text{(vowels)};\\ 9, & \text{if } \widetilde{p} \in \{\texttt{w,j}\} \quad \text{(glides)};\\ 8, & \text{if } \widetilde{p} \in \{\texttt{r,l}\} \quad \text{(liquids)};\\ 7, & \text{if } \widetilde{p} \in \{\texttt{m,n,ŋ}\} \quad \text{(nasals)};\\ 6, & \text{if } \widetilde{p} \in \{\texttt{v,ð,z}\} \quad \text{(voiced fricatives)};\\ 5, & \text{if } \widetilde{p} \in \{\texttt{s,f,h}\} \quad \text{(voiceless fricatives)};\\ 4, & \text{if } \widetilde{p} \in \{\texttt{b,d,g}\} \quad \text{(voiced stops)};\\ 3, & \text{if } \widetilde{p} \in \{\texttt{p,t,k}\} \quad \text{(voiceless stops)};\\ 1, & \text{otherwise.} \end{cases}$$

We also define the vowel set

$$\mathcal{V} \;=\; \{\texttt{a,e,i,o,u,æ}\}.$$

These mappings allow us to identify sonority peaks (nuclei) and assign each phoneme a quantitative sonority score for segmentation.

4. **Syllable Segmentation via Sonority**

For syllable boundary detection, the phoneme aligned files were processed using a rule-based algorithm rooted in the sonority sequencing principle[17]. This algorithm groups phonemes into syllables based on their sonority scores, which are derived from a predefined sonority hierarchy[17, 18]. The algorithm identifies syllable nuclei (vowels) and heuristically segments consonant clusters, splitting them at points that preserve the prosodic structure of the speech.

Given the cleaned phoneme list $\Pi_i = \{(\widetilde{p}_{i,j},\, t_{i,j}^{\text{start}},\, t_{i,j}^{\text{end}})\}_{j=1}^{n_i}$, we wish to partition the index set $\{1, \ldots, n_i\}$ into m_i contiguous intervals

$$[a_k,\, b_k] \subset \{1, \ldots, n_i\}, \quad k = 1, \ldots, m_i,$$

such that each interval corresponds to one syllable $\sigma_{i,k}$. We formally define:

$$\sigma_{i,k} \;=\; \big\{\, (\widetilde{p}_{i,j},\, t_{i,j}^{\text{start}},\, t_{i,j}^{\text{end}}) \;\big|\; j \in [a_k, b_k] \big\},$$

with the following constraints:

(a) **Nucleus Identification:** Index j is considered a *syllable nucleus* if $\widetilde{p}_{i,j} \in \mathcal{V}$ and

$$s\big(\widetilde{p}_{i,j}\big) \;>\; s\big(\widetilde{p}_{i,j-1}\big) \quad \text{and} \quad s\big(\widetilde{p}_{i,j}\big) \;>\; s\big(\widetilde{p}_{i,j+1}\big),$$

for $1 < j < n_i$. At utterance boundaries ($j = 1$ or $j = n_i$), only the available neighbor is compared.

(b) **Syllable Boundaries by Sonority Descent:** Once a nucleus j is located, we extend leftwards to find the smallest index $a_k \le j$ such that

$$s\big(\widetilde{p}_{i,a_k}\big) \;\le\; s\big(\widetilde{p}_{i,a_k+1}\big) \quad \text{or} \quad a_k = 1,$$

and extend rightwards to find the largest index $b_k \ge j$ such that

$$s\big(\widetilde{p}_{i,b_k}\big) \;\le\; s\big(\widetilde{p}_{i,b_k-1}\big) \quad \text{or} \quad b_k = n_i.$$

This ensures that each interval $[a_k, b_k]$ has a single sonority peak at its nucleus j.

(c) **Handling Consonant Clusters (Refinement):** If the interval $[a_k, b_k]$ contains a consonant-only subsequence of length greater than two (i.e., $\forall\, j' \in [a_k, b_k]$, $\widetilde{p}_{i,j'} \notin \mathcal{V}$), we split this cluster. Denote the set

$$C = \big\{\, j' \in [a_k, b_k] : \widetilde{p}_{i,j'} \notin \mathcal{V} \big\}, \quad |C| > 2.$$

We choose a midpoint $\ell = \lfloor |C|/2 \rfloor$, and partition $[a_k, b_k]$ into

$$[\, a_k,\, a_k + \ell - 1\,] \quad \text{and} \quad [\, a_k + \ell,\, b_k\,].$$

Each sub-interval must still contain at least one vowel to be labeled a valid syllable. This heuristic preserves prosodic integrity by avoiding unrealistically long, consonant-only groupings [23].

After applying these rules, we obtain the set of syllables:

$$\Sigma_i \;=\; \big\{\sigma_{i,k}\big\}_{k=1}^{m_i}.$$

Each syllable $\sigma_{i,k}$ is compactly represented by:

$$\sigma_{i,k} = \big(s_{i,k},\, e_{i,k}\big),$$

where

$$s_{i,k} = \min_{j\in[a_k,b_k]} t_{i,j}^{\text{start}}, \quad e_{i,k} = \max_{j\in[a_k,b_k]} t_{i,j}^{\text{end}}.$$

Thus $\sigma_{i,k}$ spans from the earliest phoneme onset to the latest phoneme offset within that interval.

5. **Handling Missing or Unprocessed Utterances**
   Let
   $$\begin{aligned}\mathcal{I}_{\text{input}} &= \{\texttt{speaker}_i \mid \texttt{speaker}_i \in \texttt{mfa_corpus}\}\\ \mathcal{I}_{\text{output}} &= \{\texttt{speaker}_i \mid \texttt{speaker}_i \in \texttt{mfa_output}\}.\end{aligned}$$
   Define the set of unprocessed speakers:
   $$\mathcal{U} = \mathcal{I}_{\text{input}} \;\setminus\; \mathcal{I}_{\text{output}}.$$
   For each $i \in \mathcal{U}$, we insert a dummy record
   $$\big(\texttt{phoneme_boundaries} = \varnothing,\ \texttt{syllable_boundaries} = \varnothing\big)$$
   for speaker i. This ensures that every utterance u_i remains in the final dataset, even if MFA alignment failed.
6. **Merging Aligned Boundaries into the Main Dataset**
   Let $\mathcal{R}$ denote the collection of alignment results for all utterances:
   $$\mathcal{R} \;=\; \big\{\, r_i \mid r_i = (\Pi_i,\, \Sigma_i)\big\}.$$
   We create two new columns in the main data table `df`:
   $$\texttt{df.phoneme_boundaries}[i] = \Pi_i, \quad \texttt{df.syllable_boundaries}[i] = \Sigma_i.$$
   If u_i belongs to $\mathcal{U}$, then Π_i and Σ_i are both set to $\varnothing$. This left-merge maintains utterance order and alignment integrity for all $i \in \{1, \ldots, N\}$.

- **Null Value Handling and Cleaning.**
  - All rows with null or missing transcripts were discarded.
  - Any files with unresolved audio conversion or labeling issues were excluded.
  - (`.mp4`) entries that failed conversion were removed from the final dataset.
- **Merged Dataset Statistics**
  - The final preprocessed dataset comprised 21,669 audio utterances with aligned transcriptions and emotion labels.

3.1.3 Feature Extraction

Speech emotion recognition relies heavily on the quality and variety of extracted acoustic features. In this work, we extract two broad categories of features: **temporal (frame-level) features**[20], which track moment-to-moment changes, and **non-temporal (utterance-level) features**, which give us a big-picture view of each utterance.

- **Temporal(Frame-level) Features.**
  - We develop an extraction pipeline that brings together a range of acoustic features, covering everything from basic signal properties to more complex prosodic and glottal patterns. It uses a combination of **Librosa**[21], **Parselmouth**[22], and **OpenSMILE**[23] (GeMAPS) to capture the emotional state at each frame-level moment.
  - Frame Configuration:
    - The speech signal is first downsampled to 22,050 Hz and segmented using a 32-ms frame size and an 8-ms hop size. Then on the basis of the number of frames created, each frame generates the features which are mentioned below.
  - Names of the features which are extracted are as follows:

1. Zero-Crossing Rate (ZCR).
2. Energy.
3. Entropy of Energy.
4. Spectral Features.
5. Mel-Frequency Cepstral Coefficients (MFCCs).
6. Chroma STFT & Chroma Deviation.
7. Spectral Slope.
8. Spectral Skewness.
9. Spectral Kurtosis.
10. Pitch.
11. Intensity.
12. Formants (F1, F2, F3).
13. Speaking Rate.
14. Glottal and Prosodic Features via OpenSMILE:
    (a) alphaRati_sma3
        - Ratio of spectral energy between low and high frequency bands (vocal effort).
    (b) hammarbergIndex_sma3
        - Difference in max energy between 0–2 kHz and 2–5 kHz bands.
    (c) logRelF0-H1-H2_sma3nz
        - Relative difference between the fundamental and the first two harmonics (H1–H2)
    (d) logRelF0-H1-A3_sma3nz

    - Relative difference between the fundamental and harmonic A3 (higher spectral tilt)
  (e) F1bandwidth_sma3nz
    - Bandwidth of the first formant (resonance sharpness)
  (f) F1amplitudeLogRelF0_sma3nz
    - Amplitude of F1 relative to the fundamental frequency
  (g) F2amplitudeLogRelF0_sma3nz
    - Amplitude of F2 relative to the fundamental frequency
  (h) F3amplitudeLogRelF0_sma3nz
    - Amplitude of F3 relative to the fundamental frequency

- Feature Fusion and Final Stacking
    - After extracting all individual features, they are combined into a single feature matrix. Since each utterance has a different audio length, the number of frames generated for each utterance varies. This means that each utterance will have a different number of rows in the feature matrix, while the columns represent the extracted features for each frame.

- **Non-Temporal Features**
  - Alongside the frame-wise temporal features, we also extract a rich set of global (non-temporal) voice descriptors that reflect overall voice quality, prosody, and spectral characteristics. These non-temporal features are computed once per utterance and provide summary statistics that reflect long-term patterns in pitch, intensity, harmonicity, and spectral shape. Extraction is implemented in Python using **Librosa[21] (v0.8.0)** for audio loading, **STFT**, and **RMS energy**[24], and **Parselmouth/Praat**[22] for precise pitch, formant, jitter, shimmer, and harmonicity measures. All features are assembled into a single feature vector of length 22.
  - Names of the features which are extracted are as follows:

    1. Jitter (local)
    2. Shimmer (local)
    3. HNR (Harmonics-to-Noise Ratio)
    4. Voice Tremor Index (VTI)
    5. Speech Rate
    6. Total Voiced Duration
    7. Pause Frequency
    8. Pause Duration
    9. Voicing Probability Mean
    10. Articulation Rate
    11. Mean Intensity
    12. Speaking Tempo
    13. Pitch Range
    14. Formant Ratio F2/F1
    15. Formant Ratio F3/F1
    16. Teager Energy Operator (TEO)

17. Cepstral Peak Prominence (CPP)
18. Log Energy Entropy
19. Glottal Noise Excitation (GNE)
20. Roughness
21. Pitch Confidence
22. Spectral Tilt

– All non-temporal features are extracted per sample using a unified pipeline. Invalid or failed samples were excluded. The resulting feature matrix of shape (N × 22) was compiled alongside corresponding emotion labels for classification tasks.

3.1.4 Feature Preprocessing

This section details the preprocessing, handling of anomalies, normalization, and alignment strategies for phoneme and syllable boundaries to employ the prepared feature data for further processing.

- **Feature Overview**
  - **Temporal features**: Frame-level features extracted from utterances (`shape:(21669,)`) where each element is a variable-length sequence of 51-dimensional feature vectors.
  - **Temporal labels**: Emotion annotations corresponding to each speech sample (`shape: (21669,)`).
  - **Non-temporal features**: Global-level summary characteristics such as acoustic variations, energy distribution, and statistical measures, with (`shape: (21669, 22)`).
  - **Non-temporal labels**: Emotion annotations corresponding to each speech sample (`shape: (21669,)`).
- **Sequence Length Analysis**
  - The sequence length, defined as the number of frames extracted per audio sample, was analyzed to understand its distribution across the dataset.

Table 1 Summary Statistics of Audio Sequence Lengths

Statistic Measure	***Mean***	***Median***	***Standard Deviation / Max***
Frames	278.75	239	173.75 / 4092[a]

[a]Maximum sequence length is 4092 frames.

  - Table 1 summarizes the audio sequence lengths in frames. The mean length is 278.75 frames, with a median of 239 frames, indicating a slight skew towards shorter sequences. The standard deviation is 173.75 frames, highlighting the variability in the dataset.

– Based on the distribution of sequence lengths, we chose 600 frames as the threshold for filtering out longer sequences, focusing on those that are more consistent in length.
– This variability motivated the use of sequence padding which is done in further steps. A histogram (Figure 5) illustrates the distribution of sequence lengths, overlaid with a statistical summary table.

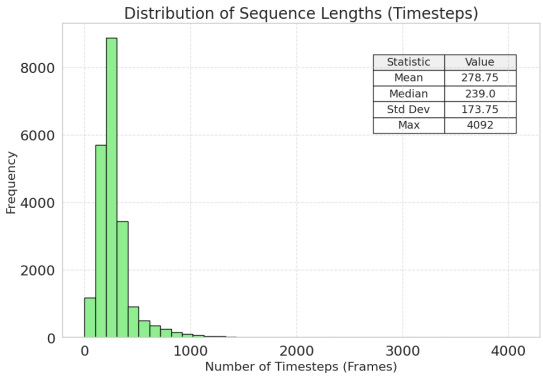


Fig. 5 Distribution of sequence lengths(Number of frames per audio)

- **Index Representation for Phoneme and Syllable Boundaries**
  To integrate phoneme/syllable boundaries into frame-wise integer index arrays, we convert continuous times into integer frame indices, which has a 600-frame input sequence. Let:

  – $F_s = 25{,}000$ Hz be the sampling frequency.
  – Δ t = 8 ms
  – T_i be the total number of frames for utterance u_i.

  Each alignment string followed the tuple format: (`'unit_label', start_time, end_time`) where `unit_label` corresponds to a phoneme or syllable, and times are in seconds.
  For each phoneme interval $(\widetilde{p}_{i,j},\, t_{i,j}^{\text{start}},\, t_{i,j}^{\text{end}}) \in \Pi_i$, define

$$f_{i,j} = \left\lfloor \frac{t_{i,j}^{\text{start}}}{\Delta\text{t}/1000} \right\rfloor, \quad g_{i,j} = \left\lfloor \frac{t_{i,j}^{\text{end}}}{\Delta\text{t}/1000} \right\rfloor.$$

  Hence, phoneme $\pi_{i,j}$ corresponds to the discrete frame interval $\left[f_{i,j},\, g_{i,j}\right]$. Similarly, each syllable $\sigma_{i,k}$ with times $(s_{i,k},\, e_{i,k})$ maps to

$$S_{i,k} = \left[\left\lfloor \frac{s_{i,k}}{0.008} \right\rfloor,\, \left\lceil \frac{e_{i,k}}{0.008} \right\rceil \right]$$

Each unique unit (phoneme/syllable) was assigned an incrementing integer ID based on order of appearance in the sequence. All frames corresponding to a unit were filled with its ID. Padding value `0` was used to denote unaligned or silent regions.

- Handling Edge Cases
    - Missing or malformed alignment strings (e.g., NaNs) were handled by returning a zero-padded array.
    - Frame indices exceeding the maximum sequence length `(max_len=600)` were clipped.
- These integer sequences represent the temporal alignment of phonemes and syllables across the fixed frame window and were used to compute learnable, multi-scale positional encodings.
- These sequences were fed into the `EmotionallySalientFrameWisePE` layer as `phoneme_positions` and `syllable_positions`. Inside this layer, we applied sinusoidal encodings with learnable scale parameters, separately for phoneme and syllable dimensions, letting the model figure out the best level of detail for encoding time.
    - `phoneme_idx_list`: a list of shape `(N, 600)` where each row corresponds to frame-wise phoneme IDs
    - `syllable_idx_list`: a list of shape `(N, 600)` with frame-wise syllable IDs

- **Handling Missing and Infinite Values**
    - Both temporal and non-temporal features were scanned for missing (NaN) or infinite values. The findings were:
        - Temporal features: Contained NaNs in some sequences.
        - Non-temporal features: Also exhibited sparse missing values.

      To address this, Mean Imputation was applied using `SimpleImputer` from `sklearn`. For temporal features, each sequence was handled independently, reshaped and imputed as needed based on dimensionality.

- **Sample Filtering and Consistency Enforcement**
  To ensure consistency across modalities and remove noisy samples:
    - A threshold of **600 timesteps** was applied to discard abnormally long sequences and to prevent outliers from changing the result.
    - Only samples meeting the following criteria were retained:
        1. Valid temporal features of shape `(T, 51)` where `T < 600`
        2. Non-temporal features of shape `(22,)`
        3. Available phoneme/syllable alignment data

      This filtering created a high-quality dataset combining frame-by-frame audio, statistical features, and linguistic boundaries.
- **Label Encoding and Train_Test Split**

  - Emotion labels were encoded using `LabelEncoder`. Stratified splitting was then applied to maintain label distribution across the training and test sets, with distribution of 80% for training and 20% for testing.

- **Feature Normalization**
  - Non-temporal Features:
    - Standardized using `StandardScaler`, fitted on the training set and applied to the test set[25].
  - Temporal Features:
    - All training sequences were stacked vertically and used to fit a `StandardScaler`[25].
    - The fitted scaler was then applied individually to each sequence.
    - This ensured that each frame-level feature dimension had zero mean and unit variance.

  This dual-scaling strategy improved convergence and stability during model training.
- **Sequence Padding and Attention Masks**
  - Since temporal sequences have variable lengths, we padded them to a fixed maximum length of **600 frames**, using `post` padding and truncation.
  - To enable attention masking in our model, binary masks were created to differentiate between real and padded tokens.

.

3.1.5 Voice-based model architecture

- Figure 6 illustrates the proposed Transformer-based architecture, which brings together several novel ideas to better understand the emotional patterns in speech. Unlike conventional models that rely solely on acoustic features, our framework integrates multi-view representations—including temporal features, global statistics, and linguistic position cues such as syllables and phonemes.
- A key innovation is the learnable salience-aware position encoding, which helps the model identify which frames in the speech signal are most emotionally significant. Combined with a hybrid attention mechanism that mixes self-attention and cross-attention, the model can effectively track how vocal tone and feature patterns shift across time in emotionally meaningful ways.
- This design allows the model not just to process raw audio, but to actively focus on the parts of speech that matter most for understanding emotion. By learning which frames carry the most emotional weight, the model can pick up on subtle shifts in tone or vocal features that often indicate changes in affective state.
- The proposed model begins with a multi-view input representation that combines temporal and non-temporal voice features. The temporal input is a matrix of shape `(600, 51)`, where 600 represents time steps and 51 is the number of acoustic features extracted per frame. These are complemented by a separate vector

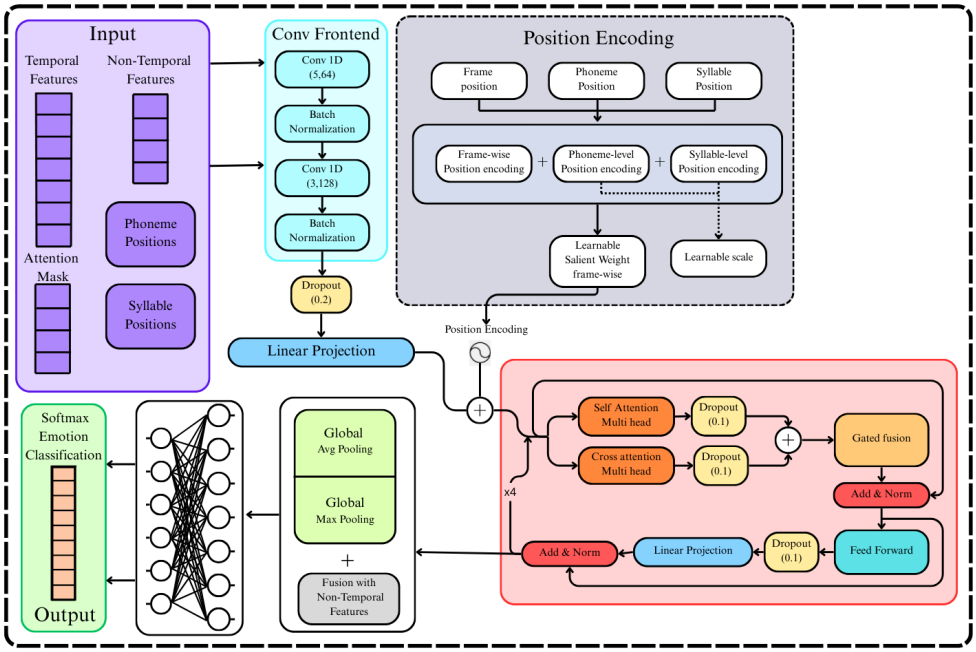


Fig. 6 Proposed Transformer-based Architecture for Emotion-Driven Voice Processing

of 22 global features. Additionally, syllable and phoneme index representation are included as positional cues. An attention mask is used to distinguish valid timesteps from padding in variable-length inputs.

This input setup is designed to provide the model with both micro-level changes in voice (via frame-wise features) and macro-level speech dynamics (via global features). Syllable and phoneme level positional encodings introduce meaningful structure, enabling the model to correlate emotional expression with prosodic boundaries in spoken language.The mathematical representation is as follows:

– **Temporal Features:**

$$\mathbf{X} \in \mathbb{R}^{B \times T \times F}$$

where:

* B: Batch size
* T: Frame-level sequence length (number of time steps)
* F: Number of extracted frame-level features per time step

– **Global Features:**

$$\mathbf{G} \in \mathbb{R}^{B \times D}$$

where D is the number of non-temporal features extracted per utterance.

– **Positional Indices:**

$$\mathbf{P}_{\text{syllable}} \in \mathbb{N}^{B\times T} \qquad \text{(Syllable level positions)}$$
$$\mathbf{P}_{\text{phoneme}} \in \mathbb{N}^{B\times T} \qquad \text{(Phoneme level positions)}$$

– **Attention Mask:**

$$\mathbf{M} \in \{0,1\}^{B\times T}$$

where $\mathbf{M}_{b,t} = 1$ indicates a valid time step and 0 denotes padding.

- The next step in the pipeline involves a 1D convolutional frontend applied to the temporal feature matrix. The model uses two successive 1D convolution layers: the first with 64 filters and a kernel size of 5, and the second with 128 filters and a kernel size of 3. Each convolutional layer is followed by batch normalization and dropout to regularize learning and improve generalization.

  These convolutional layers[14] are tasked with learning local acoustic patterns over short time windows, for example, variations in pitch or energy that may signal emotional cues like anger, sadness, or excitement. Batch normalization ensures stable learning by reducing internal covariate shift, while dropout prevents overfitting by randomly deactivating neurons during training.

  This step compresses raw acoustic data into a more meaningful representation while preserving emotionally relevant fluctuations.

  We then map the convolutional output to a latent space of $d_{\text{model}} = 128$ using a dense layer:

$$\mathbf{X}' = \mathbf{W}_{\text{proj}} * \mathbf{X}_{\text{conv}} + \mathbf{b}_{\text{proj}} \in \mathbb{R}^{B\times T\times d_{\text{model}}}$$

- **Emotionally Salient Frame-Wise Positional Encoding (ESaPE)**
  – This section introduces a custom-designed positional encoding technique tailored for emotional speech. Traditional Transformers use fixed sinusoidal encodings, but this model extends them with learnable scales for syllable and phoneme positions, allowing the model to flexibly adapt to the emotional significance of speech structures.
  Specifically, three types of encodings are combined:
    * **Standard frame-level sinusoidal positional encoding,**
    * **Syllable level positional encoding with a learnable scale θ_s**
    * **Phoneme level positional encoding with a learnable scale θ_p.**
  – These encodings are summed and then modulated by a **learnable salience vector** w_s, which highlights emotionally important time steps using ReLU activations.
  – This design helps the model pay closer attention to the emotional moments in speech, like when someone stresses a word or their voice cracks. By learning to

automatically highlight these meaningful parts through salience-aware encoding, the model becomes better at picking up on subtle emotional cues.

– It enables more precise attention, leading to emotionally contextualized representations and improved alignment between audio frames and expressive content.

– The mathematical representation is as follows:

∗ For each level, the positional encoding is computed using sinusoidal functions as:

$$\mathrm{PE}_{t,i} = \begin{cases} \sin\left(\frac{p_t}{\theta^{2i/d}}\right), & \text{if } i \text{ is even} \\ \cos\left(\frac{p_t}{\theta^{2i/d}}\right), & \text{if } i \text{ is odd} \end{cases}$$

where p_t is the position (frame index, syllable index, or phoneme index), θ is either fixed (for frame-level: $\theta = 10000$) or learnable (for syllable level: $\theta = e^{\theta_s}$, and for phoneme level: $\theta = e^{\theta_p}$) , and d is the model dimension. Therefore

$$\begin{aligned} \mathrm{PE}_{\text{frame}} &= \mathrm{PE}(pos_{frame}, 10000), \\ \mathrm{PE}_{\text{syll}} &= \mathrm{PE}(\mathbf{P}_{\text{syllable}}, \exp(\theta_s)), \\ \mathrm{PE}_{\text{phon}} &= \mathrm{PE}(\mathbf{P}_{\text{phoneme}}, \exp(\theta_p)) \end{aligned}$$

The final positional encoding is the sum of all three levels:

$$\mathrm{PE}_{\text{total}} = \mathrm{PE}_{\text{frame}} + \mathrm{PE}_{\text{syll}} + \mathrm{PE}_{\text{phon}}$$

∗ We then apply emotionally salient weighting. Let $w_s \in \mathbb{R}^{T\times 1}$ be a learnable frame-wise salience score, where T represents the number of frames. The non-negative salience scores are computed as:

$$\mathrm{PE}_{\text{mod}}(t) = \mathrm{ReLU}(w_s[t]) \cdot \mathrm{PE}_{\text{total}}(t)$$

- **Transformer Encoder Layers with Cross Attention**

– To effectively model both frame-wise temporal dynamics and utterance-level global speech characteristics, we propose a dual-attention mechanism consisting of self-attention and cross-attention layers, followed by a gated fusion strategy.

First, to support masked attention, the original attention mask is reshaped using a lambda layer. This ensures compatibility with the dimensionality requirements of the subsequent MultiHeadAttention layer by broadcasting the mask across all attention heads and time steps.
A Multi-Head Self-Attention mechanism[29] is subsequently applied to the temporal speech feature sequence X' , which is transformed into Query, Key, and Value matrices. This layer allows each frame to attend to all others, effectively capturing long-range temporal dependencies such as prosodic patterns, pitch variations, and emotional dynamics over time.

$$\text{Self Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^\top}{\sqrt{d_k}}\right)V$$

where:

* Q = Query matrix(we pass X' into this matrix in the self-attention layer)
* K = Key matrix(we pass X' into this matrix in the self-attention layer)
* V = Value matrix(we pass X' into this matrix in the self-attention layer)
* d_k = Dimension of the key vectors (used for scaling)

– Additionally, the model incorporates a Cross-Attention mechanism[28], where Temporal features are projected as Query, while Non-Temporal features are projected as Keys and Values in a multi-head attention mechanism. This step enables the model to combine high-level paralinguistic cues with local temporal dynamics, enhancing its ability to capture speaker-specific traits and the broader emotional context.
First, project the global features into a common space:

$$G' = W_g G, \quad \text{where } W_g \in \mathbb{R}^{D \times d}$$

These projected global features G' are then used as the key and value in a Multi-Head Attention module, while the query Q comes from the main input sequence X'. The cross-attention is computed as:

$$\text{Cross Attention}(X', G') = \text{softmax}\left(\frac{X'G'^\top}{\sqrt{d_k}}\right)G'$$

where d_k is the dimensionality of the keys used for scaling.

– To effectively combine the outputs from the self-attention and cross-attention modules, a gating mechanism is introduced. A dense layer with sigmoid activation learns a gate vector that determines the importance of self-attention versus cross-attention for each feature in the sequence. The model then performs a weighted fusion of the two streams using this gate. This adaptive fusion ensures that the model dynamically balances local and global information based on context.

$$\begin{aligned}\text{Fused} = \sigma(W_{\text{gate}}X') \odot \text{SelfAttention}(X') \\ + (1 - \sigma(W_{\text{gate}}X')) \odot \text{CrossAttention}(X', G')\end{aligned}$$

– To maintain stability during training and ensure efficient learning, the fused representation is added back to the original input X' using a residual connection, followed by layer normalization. This technique helps keep the gradient flow smooth across layers and prevents the network from drifting too far from the original input signal. By combining the new attention-derived features with the original input, the model balances learning from both sources—avoiding over-reliance on either—and builds a more robust foundation for deeper learning. The complete mathematical formulation of this step is given below:

$$\mathbf{x} = \text{LayerNorm}(\text{Fused} + \mathbf{X}')$$

where:

    - Fused is the fused representation from the attention mechanisms,
    - X' is the original input.
    - LayerNorm denotes the layer normalization operation, which normalizes the input across the features.

  – Finally, the model passes the fused attention output through a feedforward neural network[26] that includes a ReLU activation[27], dropout for regularization, and a projection layer to return the data to its original dimensionality. This step gives the model more expressive power by introducing non-linear transformations, helping it better capture the complex patterns in emotional speech. In essence, this feedforward layer works like a refinement stage, reshaping the contextualized features into a space that's more useful and discriminative for accurately recognizing emotions.

$$\text{FFN}(x) = W_2 \left(\text{ReLU} \left(W_1 x + b_1\right)\right) + b_2$$

  – The feedforward network output $FFN(x)$ is added to the result of the previous addition and normalization step x via a residual connection and then passed through a layer normalization layer.

$$\mathbf{x} = \text{LayerNorm}(\text{FFN}(\mathbf{x}) + \mathbf{x})$$

- After the transformer layers, the model aggregates information from all time steps through global pooling. It performs Global Average Pooling (AvgPool) and Global Max Pooling (MaxPool) across the temporal dimension, reducing the output sequence to a single vector representing the entire speech segment. These pooled representations are concatenated with a projection of the Non-Temporal features.
  The purpose of this fusion is to merge temporal details (extracted from the transformer layers) with global prosodic features. These patterns reflect broader speech characteristics that unfold over the course of an entire utterance. By combining moment-to-moment changes with the overall tone and style of the speech, the model can make more accurate and nuanced predictions that consider both the fine details and the bigger picture.

$$X_{\text{Pooled}} = [\text{AvgPool}(x), \text{MaxPool}(x)]$$

$$F_{\text{global}} = [X_{\text{Pooled}}, \text{ReLU}(W_p, G)]$$

- After the feature fusion, the final layer of the model is a fully connected classification head, consisting of multiple dense layers with ReLU activations followed by dropout for regularization.
  The final dense layer outputs predictions across 9 emotion classes using the Softmax activation to convert logits into probabilities.
  The primary function of this classification head is to map the fused feature representations into specific emotion categories (e.g., happy, sad, angry, etc.). By using multiple dense layers, the model can learn complex non-linear mappings between

the high-dimensional features and the final classification space.

$$\hat{y}_{\text{voice}} = \text{Softmax}\left(\text{ReLU}\left(W_4 \cdot \text{ReLU}\left(W_3 \cdot F_{\text{global}} + b_3\right) + b_4\right) + b_5\right)$$

$$\hat{y}_{\text{voice}} \in \mathbb{R}^9$$

- **Focal Loss with Label Smoothing**
  - The model uses Focal Loss with label smoothing to optimize its performance, particularly for datasets where certain emotion classes may be underrepresented.
  - Label smoothing transforms the true label y from one-hot to a softer label[31]

    $$y_{\text{smooth}} = (1 - \epsilon)y + \frac{\epsilon}{C}$$

    where:
    - $\epsilon = 0.2$ is the smoothing parameter, which softens the one-hot labels, making the model less confident and helping to avoid overfitting.
    - C is the number of classes.
  - Focal Loss is designed to down-weight easy examples and focus on hard-to-classify instances, especially in the case of class imbalance[30]. The formula for focal loss is:

    $$\mathcal{L}_{\text{focal}} = -\alpha(1 - \hat{y})^{\gamma} y \log(\hat{y})$$

    where:
    - The balancing factor $\alpha = 0.8$ adjusts the importance of different classes.
    - The focusing parameter $\gamma = 3.0$ down-weights easy-to-classify examples.
  - This loss function addresses class imbalance, ensuring that the model is more sensitive to underrepresented emotions[30, 31]. By preventing the model from becoming overconfident in certain classes.

- **Training Strategy**
  - The model adopts the Adam optimizer[32] with a learning rate that begins at 1×10^{-4}, followed by cosine decay[33]. This dynamic learning rate scheduling helps in avoiding local minima and ensures that the model can escape flat regions of the loss landscape. The parameters for cosine decay are $T_0 = 1000$ and $T_{\text{mul}} = 2$, allowing the learning rate to reset periodically, providing the model with fresh learning opportunities.

3.2 Text-based model

3.2.1 Data collection

Emotion recognition models really only get as good as the data they're trained on. That's why we picked datasets that cover a wide range of emotional expressions, conversational settings, and language styles. By using this diverse mix, we give our text-based emotion recognizer the solid foundation it needs to learn and to be evaluated fairly. Below are the specific datasets we included in our work.

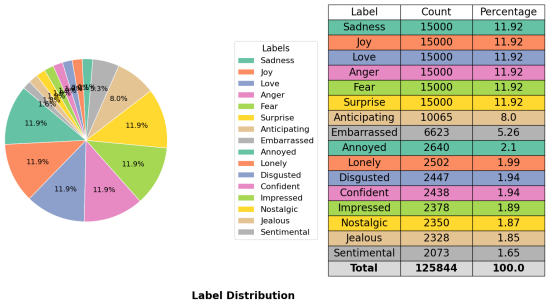
- Empathic Dialogues
  - EmpatheticDialogues is a large-scale dataset consisting of 76,668 open-domain dialogues grounded in emotional situations, designed to encourage empathetic responses. Each dialogue was initiated with a situation reflecting one of 32 emotion labels (e.g., pride, disappointment, excitement, guilt), and crowd workers responded empathetically to the speaker [34].
  - The dataset was collected via Amazon Mechanical Turk and emphasizes emotional alignment between the speaker and listener. It is widely used for emotion recognition, empathetic response generation, and affective dialogue modeling.
- Dair AI emotion dataset
  - The Dair AI dataset consists of 4,16,809 dialogues from human-human interactions, specifically designed for AI-driven conversational agents. It includes a diverse array of emotional expressions, from positive emotions like excitement and joy to negative ones like anger and fear. This dataset also includes contextual metadata that provides insights into the speaker's emotional state during the conversation, such as speech rate, pauses, and tone variations.
  - This dataset is unique in its inclusion of multimodal emotional cues (text, tone, and prosody) alongside textual data.

The datasets were merged, resulting in a comprehensive training set that contains over 4,93,430 labeled text samples.

3.2.2 Data Preprocessing

- **Raw Text Cleaning**
  1. Lowercasing
  2. Remove non-alphanumeric characters (regex `[â-zA-Z0-9 ]`)
  3. Strip punctuation (via str.translate)
  4. Remove digits `(re.sub(r'ṫ', ''))`
  5. Whitespace normalization
- **Filtering the dataset**
  - The merged dataset exhibited a significant class imbalance, with approximately 30% of the total samples labeled as `joy`, while the remaining 70% were distributed across the other 37 emotion labels.
  - To mitigate this imbalance and ensure a more equitable class distribution, a threshold-based sampling strategy was employed.
  - Specifically, a maximum cap of 15,000 samples per emotion class was enforced. This approach aimed to prevent dominant classes from biasing the model while preserving sufficient data diversity for minority classes.
  - After applying this filtering approach, the final dataset consisted of 125,844 samples, which were used for training the model.
- **Label classification**

– In the merged dataset, an initial set of 38 distinct emotion labels was identified across the combined corpora. To address semantic redundancy and enhance label consistency, these were systematically consolidated into 16 core emotion categories by mapping semantically similar labels. For instance, terms such as 'sad', 'devastated', and 'disappointed' were grouped under the unified label 'sadness'. Likewise, 'joyful', 'content', 'proud', and 'grateful' were merged into 'joy'; 'caring', 'trusting', and 'faithful' were classified as 'love'; and 'afraid', 'terrified', 'anxious', and 'apprehensive' were grouped into 'fear'. Additional mappings included 'surprised' to 'surprise', 'ashamed' and 'guilty' to 'embarrassed', 'hopeful', 'excited', and 'prepared' to 'anticipating', and 'angry' and 'furious' to 'anger'.
– After applying the filtering and label classification procedures, the final set was consolidated into 16 distinct emotions as shown in Figure 7. These standardized labels include { `sadness, joy, love, anger, fear, surprise, anticipating, embarrassed, annoyed, lonely, disgusted, confident, impressed, nostalgic, jealous, and sentimental`}. The final label distribution is illustrated in Figure 7.



Label	Count	Percentage
Sadness	15000	11.92
Joy	15000	11.92
Love	15000	11.92
Anger	15000	11.92
Fear	15000	11.92
Surprise	15000	11.92
Anticipating	10065	8.0
Embarrassed	6623	5.26
Annoyed	2640	2.1
Lonely	2502	1.99
Disgusted	2447	1.94
Confident	2438	1.94
Impressed	2378	1.89
Nostalgic	2350	1.87
Jealous	2328	1.85
Sentimental	2073	1.65
Total	**125844**	**100.0**

Fig. 7 Class distribution in text-based model across multiple emotion categories

- **Tokenization & Encoding**
  – With clean text in hand, we tokenize and encode using Hugging Face's DistilBERT tokenizer `AutoTokenizer.from_pretrained("distilbert-base-uncased")`[35]. All sentences are padded or truncated to a fixed length of 128 tokens, yielding input_ids and attention_mask arrays.
- **Label Encoding and Train_Test Split**
  – Emotion labels were encoded using `LabelEncoder`. Stratified splitting was then applied to maintain label distribution across the training and test sets, with distribution of 80% for training and 20% for testing.

3.2.3 Text-based model architecture

- At the core of this architecture lies the pretrained transformer model `DistilBERT`[36], which serves as a powerful feature extractor. It takes in tokenized

textual input and captures contextual embeddings for each token in the sequence. This significantly reduces the need for manual feature engineering and allows the model to understand the semantic nuances of the input sentences.

- On top of the frozen or fine-tuned DistilBERT base, a custom classification layer is added. This consists of a single Dense layer with a softmax activation function, which maps the [CLS] token embedding (i.e., the first token's output representation from DistilBERT) to a fixed number of emotion classes (equal to the number of harmonized labels in the dataset).
- The entire architecture is defined using TensorFlow's `tf.keras.Model` subclassing API. This provides flexibility and transparency in customizing the forward pass—specifically, selecting only the [CLS] token output for classification.
- The model is compiled with the `SparseCategoricalCrossentropy` loss function[37], which is suitable for integer-encoded multi-class classification tasks. An `Adam` optimizer with a learning rate of `1e-5` is used for stable and gradual fine-tuning of the pretrained weights[32].
- The model is trained using TensorFlow Datasets, which wrap tokenized and labeled input into efficient batches. A batch size of 128 is chosen, and the model is trained for 20 epochs, with validation performed on a held-out test set at each epoch to monitor generalization.
- The text-based model produces an emotion probability vector

$$\hat{y}_{\text{voice}} \in \mathbb{R}^{16}$$

, where each element corresponds to the predicted likelihood of one of the 16 predefined emotion categories.

3.3 Late Fusion Stratergy

Now that both the text-based and voice-based classifiers have been independently trained on their respective datasets, we employ a late fusion framework to merge their decision-level outputs.

- **Unimodal Outputs**
  - The voice-based model outputs a 9-dimensional softmax probability distribution:

$$P_{\text{voice}} \in \mathbb{R}^{9},$$

    where each dimension corresponds to one of the predefined emotion categories.
  - The text-based model outputs a 16-dimensional softmax probability distribution:

$$P_{\text{text}} \in \mathbb{R}^{16},$$

    where each dimension corresponds to one of the predefined emotion categories.
- **Text-Based Projection into Voice-Emotion Space**

– To reconcile the differing label spaces, we compute a projection matrix

$$\mathbf{N} \in \mathbb{R}^{9\times 16}$$

by measuring pairwise cosine similarities between the `[CLS]` embeddings of text and voice labels, and applying a temperature-scaled softmax normalization.
– This transformation maps the 16-dimensional text probability vector into the 9-dimensional voice label space, ensuring that semantically related labels receive higher cross-modal correspondence.
– The projected text probabilities are obtained as

$$P_{\text{text}\to\text{voice}} = \left(N P_{\text{text}}^{\top}\right)^{\top},$$

yielding a 9-dimensional vector aligned with the voice label space.

- **Weighted Probability Averaging**
  – We perform a grid search over the fusion weight $w \in [0, 1]$ and define

$$P_{\text{voice}} \in \mathbb{R}^9, \quad P_{\text{text}\to\text{voice}} \in \mathbb{R}^9.$$

$$P_{\text{final}}(w) \;=\; w\,P_{\text{voice}} \;+\; \big(1-w\big)\,P_{\text{text}\to\text{voice}}, \quad w \in [0,1].$$

The optimal weight w^* (e.g., 0.27) is selected based on maximum validation accuracy, striking a balance between the modalities.

- **Meta-Learner Ensemble**
  – This method allows us to combine the high-level meanings and acoustic features captured by models trained separately, while still preserving the unique strengths of each model.
  – We construct an 18-dimensional *meta-feature vector* for each utterance by concatenating the two 9-dimensional vectors:

$$\mathbf{x}_{\text{meta}} = [P_{\text{voice}}; P_{\text{text}\to\text{voice}}] \in \mathbb{R}^{18}.$$

This vector brings together confidence scores from different modalities into a single representation, making it easier to perform strong cross-modal reasoning.
  – A *Random Forest classifier* is trained on these 18-dimensional vectors to act as the final decision-making module[38]. It learns to recognize complex patterns within the combined probability distributions, which may not be easily separated in a linear way.

$$\hat{y} = \text{MetaClassifier}(\mathbf{x}_{\text{meta}}),$$

where $\hat{y}$ is the predicted emotion label.
  – The meta-learner uses ensemble-based voting over decision trees, making it ideal for capturing complex, nonlinear interactions between features and managing moderately imbalanced classes.

4 Results

4.1 Voice-based model

Our voice-based system uses a custom Transformer architecture with Emotionally Salient Frame-wise Positional Encoding (ESaPE), syllable and phoneme level temporal encoding, and cross-attention with global prosodic features. After training for 60 epochs with a batch size of 16, the model achieved an overall accuracy of 60.71% on the test set.

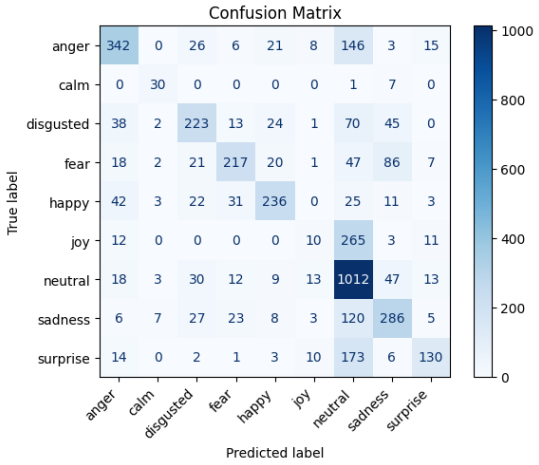


Fig. 8 Confusion Matrix of Voice-Based Model

The Figure 8 highlights the model's ability to accurately classify emotions such as 'neutral' and 'anger', which show high true positive rates (e.g., 1012 correct predictions for 'neutral' and 342 for 'anger'). However, it also reveals significant misclassifications, especially among emotionally similar classes. For instance, 'fear' and 'disgusted' are frequently misclassified with one another and with 'neutral', suggesting overlapping acoustic features or model limitations in distinguishing subtle vocal expressions. This pattern points to the need for improved feature extraction or model architectures better tuned for emotional nuance in speech.

The high confusion between certain categories, such as 'surprise' being often mistaken for 'neutral' (173 instances) or 'joy' being confused with 'neutral' and 'anger', underscores the complexity of emotion representation in vocal signals. These findings suggest that current models may benefit from incorporating contextual or semantic cues alongside raw audio features.

The figure 9 shows that model exhibits the highest performance in detecting the neutral emotion, achieving a recall of 0.87 and an F1 score of 0.67, indicating strong sensitivity to neutral speech and a balanced precision-recall tradeoff.

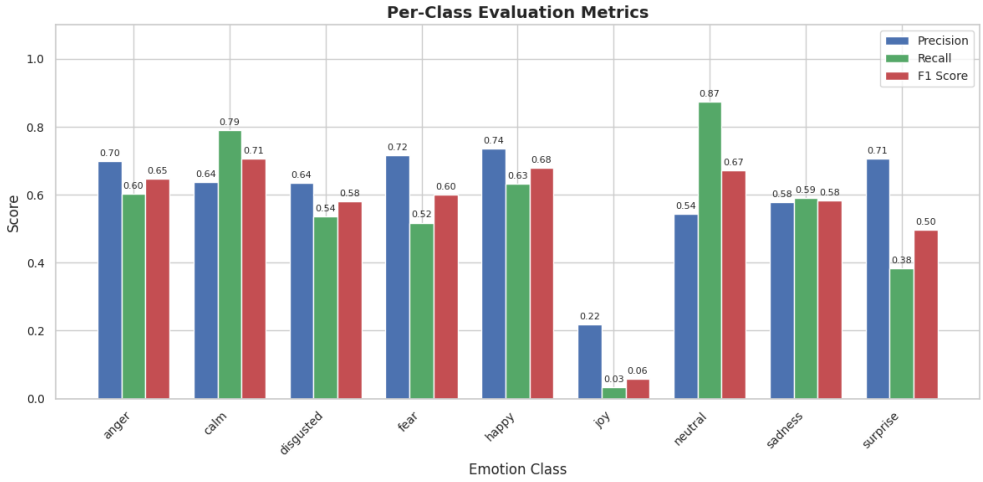
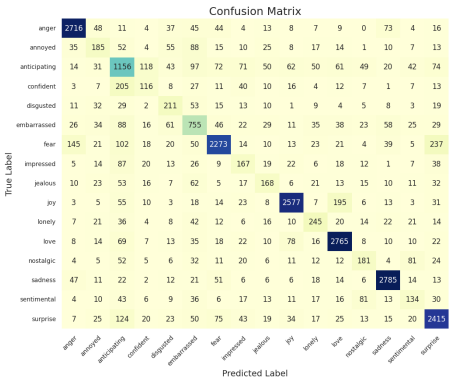


Fig. 9 Classwise Performance

The joy class shows the weakest performance with a recall of 0.03 and F1 score of 0.06, suggesting significant difficulty in accurately identifying joyful expressions, potentially due to class imbalance or overlapping acoustic features with other emotions.

4.2 Text-based model

A DistilBERT-based text classifier fine-tuned using the [CLS] token representation, trained for 20 epochs with Adam optimizer and sparse categorical crossentropy on labeled emotion datasets gives the overall accuracy of 74.89%.



Precision, Recall, F1-score per Emotion Class

	precision	recall	f1-score
anger	0.89	0.89	0.89
annoyed	0.38	0.34	0.36
anticipating	0.53	0.58	0.55
confident	0.32	0.24	0.27
disgusted	0.4	0.5	0.44
embarrassed	0.53	0.58	0.55
fear	0.85	0.76	0.8
impressed	0.34	0.36	0.35
jealous	0.41	0.36	0.38
joy	0.9	0.87	0.88
lonely	0.49	0.49	0.49
love	0.85	0.89	0.87
nostalgic	0.43	0.39	0.41
sadness	0.9	0.92	0.91
sentimental	0.34	0.3	0.32
surprise	0.8	0.83	0.81

Fig. 10 Performance Evaluation of Text-Based Model

The confusion matrix in Figure 10 reveals that the model demonstrates strong classification performance on high-resource emotions such as anger, joy, love, and sadness, with precision and recall values exceeding 85% for each. This indicates a reliable detection capability for these core emotions. However, emotions such as confident, impressed, and sentimental exhibit considerably lower performance metrics, with F1-scores dropping to as low as 0.27–0.35. This disparity likely stems from class imbalance or the semantic overlap among nuanced emotions, which introduces ambiguity in distinguishing between them, especially in textual contexts where tone and prosody are absent.

The aggregated metrics from the classification report indicate that while the model performs well on dominant classes, it struggles with less frequent or subjectively complex emotions. For instance, the class anticipating shows moderate performance (F1-score = 0.55), while nostalgic and sentimental hover near the lower spectrum. These results suggest that the model may benefit from enhanced feature representations or auxiliary data to capture subtle emotional cues.

4.3 Fusion of Voice and Text-Based Model

Both the voice-based and text-based models independently capture complementary emotional cues from speech and textual content, respectively. While each model performs reasonably well on its own (Table 2), they exhibit limitations when used in isolation due to the inherent ambiguity and noise in individual modalities. To leverage the strengths of both, we employ late fusion techniques, combining their outputs to improve overall emotion recognition accuracy.

Table 2 Comparison of Key Metrics for Voice-Based and Text-Based Model Before Fusion

Model	Accuracy (%)	F1-Score	Precision	Recall
Voice-Based Model	60.70	0.58	0.60	0.61
Text-Based Model	74.89	0.75	0.75	0.75

Note: Metrics are averaged over all emotion classes.

We explore two late fusion strategies to combine the predictions from the voice-based and text-based models:

1. **Weighted Probability Averaging:**
   The predicted class probabilities from each model are combined using empirically determined weights. Figure 11 illustrates the impact of varying the fusion weight (w) between voice and text modalities on the final prediction accuracy of a multimodal emotion recognition model. The best results are obtained at `w = 0.27`, suggesting that [voice/text]-dominant fusion produces the best classification accuracy.
2. **Meta-Learner Ensemble:**
   The output probabilities from each model are concatenated and fed into a meta-classifier (e.g., random classifier) to learn optimal combination rules.We used a

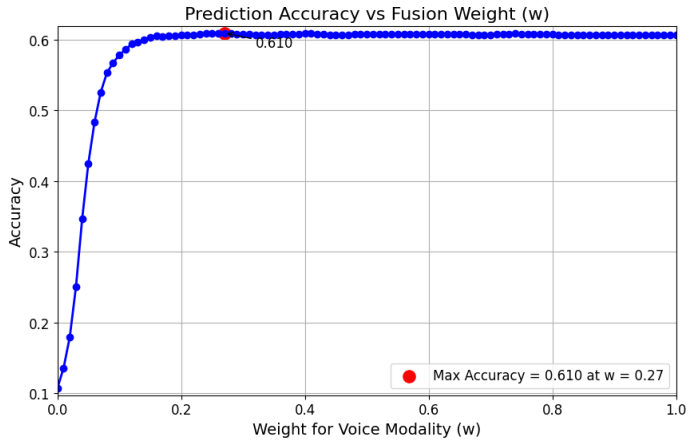


Fig. 11 Accuracy vs. fusion weight w for combining voice and text modalities. The red point marks the optimal weight achieving the highest accuracy.

Random Forest classifier to make the meta-learner. It was made to find a good balance between performance and generalization. It had 200 decision trees and could only go as deep as 15 to lower the chance of overfitting. The square root strategy (`max_features='sqrt'`) was used to figure out how many features to look at at each split. To make it easier to generalize, we needed at least two samples to split an internal node and at least two samples at each leaf. To make the training process more efficient, it used all of the available CPU cores. It was also made reproducible by setting a fixed random seed. This setup helped the meta-classifier learn how to combine the outputs of the base models, which in turn improved the performance of emotion recognition.

The performance of both late fusion strategies, along with the individual models, is summarized in Table 3. It is evident that combining the modalities through late fusion leads to a noticeable improvement in emotion recognition accuracy and other key metrics.

Table 3 Comparison of Unimodal and Multimodal approaches

Model	**Method**	**Accuracy (%)**	**F1-Score**	**Precision**	**Recall**
Voice-Based Model	-	60.70	0.58	0.60	0.61
Text-Based Model	-	74.89	0.75	0.75	0.75
Late Fusion	Weighted Probability Avg.	61.00	0.58	0.60	0.60
	Meta-Learner Ensemble	61.32	0.60	0.61	0.61

Note: Metrics are averaged over all emotion classes.

5 Conclusion

In this work, we proposed a multimodal emotion recognition system that integrates voice-based and text-based cues to enhance the accuracy and reliability of emotion classification. We first developed independent models for each modality, leveraging temporal and non-temporal speech features for the voice-based model and contextual embeddings for the text-based model. While both models demonstrated competitive standalone performance, they exhibited limitations when operating in isolation due to modality-specific ambiguities and noise.

To fix this, we used two late fusion methods—weighted averaging and meta-classifier stacking—to combine the predictions of both models at the decision level. Our results showed that late fusion made all of the most important performance metrics much better. The meta-classifier stacking method did better than the individual models and the weighted averaging approach, with a score of 61.32%. This shows that learning to combine predictions from different modalities in a flexible way is helpful for picking up on complicated emotional cues. This work lays a solid basis for future research in this area and emphasizes the significance of multimodal integration in emotion recognition systems. We intend to go beyond decision-level integration with our fusion strategies in subsequent work. In particular, we intend to investigate hierarchical fusion methods that integrate intermediate-layer feature interactions and joint representation learning via cross-modal attention mechanisms. Furthermore, experimenting with emotionally salient positional encoding and incorporating temporal structures at the phoneme and syllable levels can enhance the model's sensitivity to subtle emotional changes. A promising approach to attaining deeper modality alignment is to look into transformer-based architectures with shared decoders and modality-specific encoders.

By advancing toward more sophisticated and semantically aware fusion methods, we envision building emotion recognition systems that are not only more accurate but also more robust and adaptable across diverse real-world contexts. Subsequent research could investigate dynamic fusion approaches that learn and adjust according to context or individual-specific emotional trends. Moreover, cross-lingual and cross-cultural affective variations that are integrated could further enrich the system's global applicability as well as inclusivity.

6 Declaration

6.1 Ethical Approval

not applicable

6.2 Funding

not appicable

6.3 Availability of data and materials

not applicable

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